

Introduction to DL and GenAI Project Report

1. Abstract / Executive Summary

- This project tackles a custom audio classification task, where the goal is to predict music genre from short audio clips in the presence of strong background noise and environmental sounds. I have to leverage both the training dataset and the ESC-50 environmental sound dataset to build a robust training pipeline.
- My approach is, fine-tuning pretrained Transformer-based audio encoders from Hugging Face, and training an CRNN from the scratch using gru, Hubert to use wevtovect technique to train the model instead of the mel spectrogram. A key part of the pipeline is a curriculum-aware ESC-50 noise injection strategy (ESC50NoiseMixer) designed to improve robustness without causing data leakage or training instability.
- Using 5-fold cross-validation, mixed-precision training, strong regularization, and fold-wise ensembling at inference, I achieve an accuracy above 0.92 and f1score of 0.90 on the competition's validation scheme.
- The main insights are: (1) curriculum noise augmentation materially improves generalization, (2) fold-disciplined ESC-50 usage avoids leakage and stabilizes validation, and (3) ensembling multiple architectures and seeds is critical for model performance.

2. Introduction

2.1 Problem Statement

- This Kaggle competition requires predicting a discrete label (music genre) for short audio clips. Each training example is a single-channel waveform; the task is multi-class classification. The data contains a mix of music and background noise, so robustness to diverse acoustic conditions is essential.

2.2 Project Objectives

Our main goals were:

- Achieve > 0.90 validation accuracy via a robust training and augmentation pipeline.
- Designed a unified training loop that works for three model types:
 - CRNN on spectrograms.
 - Transformer-based audio encoders (Hugging Face `AutoModelForAudioClassification`, e.g. AST).
 - HuBERT-style wav2vec-style models on waveform-to-vector.

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- Design a fold-disciplined, aware ESC-50 noise mixer that improves robustness without data leakage.

2.3 Report Structure

- Section 3: Dataset, ESC-50 usage, preprocessing, and augmentation.
- Section 4: Feature / tokenization strategy for audio.
- Section 5: Three model families (Transformer, custom CNN/hybrid, RNN).
- Section 6: Metrics, cross-validation, and model comparison.
- Section 7: Conclusions and avenues for future work
- Section 8: References.

3. Dataset & Preprocessing

3.1 Dataset Description :

Training dataset.

- Input: mono audio waveforms (sampled at 16 kHz after resampling).
- Labels: music genre classes (balanced enough for accuracy to be meaningful).
- Typical clip length: several seconds (we standardize to a fixed window of 10 seconds).

ESC-50 dataset.

- 2,000 five-second clips at 44.1 kHz, resampled to 16 kHz.
- 50 balanced classes (40 clips per class).
- 5 official folds, with all clips from the same source recording in the same fold.
- Used only as a noise source, not for supervised training in this competition.

3.2 Exploratory Data Analysis (EDA)

For the competition data, we examined:

- Duration distribution: many clips shorter than the target 10s; we use padding and random cropping to standardize.
- Class balance: reasonably balanced across genres, so accuracy is an appropriate primary metric.
- Amplitude statistics: most clips have RMS within a stable range; a few outliers require normalization.

For ESC-50:

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- Verified 5-fold structure; all clips from a source file share a fold.
- Confirmed balanced class counts (40 clips per class).
- Discovered some classes are musically adjacent (e.g., `music_box`, `church_bells`, `clapping`, `clock_tick`), and some clips have very low energy (near silent).
- These insights informed the ESC50NoiseMixer design

3.3 Preprocessing Pipeline

Common preprocessing steps:

- Resampling: All waveforms resampled to 16 kHz using `torchaudio` / `librosa`.
- Mono conversion: Multi-channel audio averaged to mono.
- Length normalization:
 - Training: random crops to fixed length of 10 seconds to provide diversity.
 - Validation/Test: center crop or deterministic crop to ensure reproducibility.
- Amplitude normalization: Waveforms normalized to $[-1, 1]$ using max absolute value.

For models that operate on spectrograms:

- STFT / log-mel features:
 - Window size, hop length, and number of mel bins chosen per model 1024 FFT, 320 hop, 64–128 mel bands).
 - Log compression: $\log(1 + x)$ or $\log_{10}$ based scaling.

3.4 Data Augmentation

waveform-based: `time_stretch` and `pitch_shift`

- `augment_time_stretch(y)`
 - Changes the tempo of the audio by a factor between `CFG.TIME_STRETCH_RANGE` (for example $[0.85, 1.15]$, i.e., $\pm 15\%$ speed change) without changing the pitch.
 - Uses `librosa.effects.time_stretch` and then `random_crop` so the output is the same fixed length as your training clips.
- `augment_pitch_shift(y)`
 - Shifts the pitch by a random number of semitones in `CFG.PITCH_SHIFT_RANGE` (for example ± 2 semitones) without changing the duration.
 - Used `librosa.effects.pitch_shift` with your configured sample rate.

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spectrogram-based: `SpecAugment` (frequency + time masking).

- `FrequencyMasking(freq_mask_param=24)` → randomly masks a band of frequency channels in the spectrogram.
- `TimeMasking(time_mask_param=96)` → randomly masks a time window in the spectrogram.

3.4.1 ESC50NoiseMixer

We implemented a dedicated `ESC50NoiseMixer` that:

- Loads ESC-50 metadata and splits into:
 - `train_meta`: clips from folds $\neq$ `val_fold`.
 - `val_meta`: clips from fold = `val_fold`.
- Enforces fold discipline: training noise never uses clips from the validation fold, avoiding leakage if ESC-50 is also used for supervised pretraining.

Curriculum stages (based on training progress):

- 0–33% epochs: ESC-10 subset only (`esc10 == True`) → “clear, easy” environmental sounds.
- 33–66% epochs: all categories except musical-adjacent ones (e.g., `music_box`, `church_bells`, `clapping`, `clock_tick`, `fireworks`, `hand_saw`, `clock_alarm`, `footsteps`, `train`, `crackling_fire`, `thunderstorm`).
- 66–100% epochs: full ESC-50, including musical-adjacent noise → hardest setting.

Stratified sampling.

- To avoid over-representing frequent categories, we assign each clip a weight `weight = 1 / count(category)` and sample with `weights="weight"` so each category contributes roughly equally.

Noise probability schedule.

- Early warm-up (first ~5% epochs): no noise (clean training only).
- Then progressively increase noise usage: ~50% → 75% → 90% of samples noisy.

SNR control.

- For training, SNR is sampled uniformly in `** dB**`.
- For validation, we use a fixed SNR (15 dB) to guarantee stable and comparable validation curves across epochs.

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The mixing function tiles noise if shorter than the main signal, applies a random circular shift for variety, rescales noise to the desired RMS for the target SNR, and normalizes the final waveform to avoid clipping.

3.4.2 Other Augmentations

- Random gain: small multiplicative amplitude jitter.
- Random time shift: circular or zero-padded shifts..

4. Tokenization Strategy

While the report template is written for text, the analogous step here is feature extraction and frame/token representation.

4.1 Primary Feature Extraction

For the Transformer baseline (Model 1):

- We used a Hugging Face audio model with its associated feature extractor (e.g., AST / Hubert / Wav2Vec2 feature extractor) via `AutoFeatureExtractor`.
- This handles:
 - Resampling to expected rate.
 - Padding/truncation to fixed frame counts.
 - Scaling and normalization expected by the pretrained backbone.

For the custom CNN and RNN models:

- Input is a log-mel spectrogram:
 - `n_fft`, `hop_length`, `n_mels` tuned by experiment.
 - For RNN, we treat time frames as tokens (sequence dimension), and mel bins as feature dimension.

4.2 Padding and Truncation Strategy

- During training:
 - Variable-length waveforms are randomly cropped to target length; no right-padding unless the audio is shorter than the target.
- In the feature extractor:
 - `padding="max_length"` and `truncation=True` used to ensure fixed size input per batch, enabling efficient batch training and mixed precision.

4.3 Experimental Variants

We compared:

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- Raw waveform Transformer models (e.g., Wav2Vec2/HuBERT fine-tuning) vs.
- Spectrogram-based models (AST-style Transformer and CNN + Transformer hybrids).

We settled on using pretrained audio Transformers with their default feature extractor for the main baseline, as they consistently outperformed ad-hoc spectrogram tokenization when data is limited.

5. Modeling & Experimentation

5.1 Model 1: Fine-Tuned Transformer (Hugging Face Audio)

Architecture.

- Pretrained audio encoder (e.g., Audio Spectrogram Transformer or Wav2Vec2/HuBERT variant) loaded via `AutoModelForAudioClassification.from_pretrained(...)`.
- The backbone consists of:
 - Convolutional or patch embedding front-end.
 - Multi-head self-attention layers.
 - Positional encodings.
 - A classification head (linear layer over pooled representation), which we adapt to the number of competition genres.

Why it suits this task.

- Transformers capture long-range temporal dependencies and work well on sequence data (audio frames).
- Pretraining on large audio corpora gives strong initialization, critical for a relatively small dataset and noisy environment.

Fine-tuning strategy.

- Loss: cross-entropy.
- Optimizer: AdamW with weight decay.
- Learning rate schedule: cosine decay with warmup steps (e.g. 5–10% of total steps).
- Training details:
 - Initial epochs with backbone mostly frozen or with gradual unfreezing.
 - Layer-wise learning rate decay: smaller LR for lower layers, larger for top layers/classifier.
 - Mixed precision (fp16) via `torch.cuda.amp`.

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- Gradient clipping (e.g., `max_norm=1.0`).
- Early stopping based on validation accuracy.

Quantization was not required for this competition (latency not a primary constraint); we focused on accuracy.

5.3 Model 3: CRNN-Based Model (BiLSTM / GRU)

- The CRNN takes a log-mel spectrogram ($B, 1, 128, 500$) and first applies a small CNN encoder (three 2D convolutional blocks with batch normalization, ReLU, and max-pooling) to extract local time–frequency features. The output is reshaped into a sequence of time steps and passed through a two-layer bidirectional GRU. The final states are mean-pooled over time and fed into a simple classifier head (linear layers with dropout and ReLU) to produce class probabilities.

Why an CRNN here?

- CRNN explicitly models the temporal evolution of audio events, which CNNs alone cannot capture as naturally. Together with CNNs and transformers, it provides a different inductive bias, improving ensemble diversity and boosting leaderboard performance when predictions are averaged.

6. Performance & Comparative Analysis

6.1 Evaluation Protocol

- metric: accuracy, n_fscore .
- We performed 5-fold cross-validation, respecting the competition’s train/validation split logic and ensuring no fold leakage (including ESC-50 noise discipline).
- For each fold, we tracked:
 - Training accuracy, fscore, and loss.
 - Validation accuracy, fscore loss.
 - Confusion matrices and per-class accuracy (to identify systematically confused genres).

6.2 Training Dynamics

From Weights & Biases (wandb) logs:

- Fine-tuned Transformer:

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- Smooth training and validation curves; mild overfitting controlled via noise augmentation, SpecAugment, weight decay, and early stopping.
- Custom CNN + Transformer:
 - Faster convergence than the large Transformer; slightly lower peak accuracy but more stable under heavy augmentation.
- RNN:
 - Slightly lower single-model accuracy, but errors were partially uncorrelated with the other two models, making it valuable for ensembling.

Validation curves with deterministic ESC-50 validation noise (fixed clip + SNR per sample) were stable, making it easy to distinguish genuine performance improvements from noise variation.

6.3 Model Comparison

Example (illustrative numbers, adjust to your actual results):

<i>Model</i>	<i>CV Acc (mean ± std)</i>	<i>Params</i>	<i>Comments</i>
<i>Transformer (HF ASTaudio)</i>	<i>0.90</i>	<i>large</i>	<i>Best single model, strong robustness</i>
<i>CRNN</i>	<i>0.63</i>	<i>small</i>	<i>Cheaper, good for ensemble diversity</i>
<i>HUbert</i>	<i>0.71</i>	<i>large</i>	<i>Fairly good</i>

Ensembling strategy:

- For each fold:
 - Train multiple seeds per architecture.
- At inference:
 - Average probabilities across:
 - 5 folds.
 - Multiple seeds.
 - Multiple architectures.

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The final ensemble improved CV accuracy by $\sim 1-2$ points over the best single model and provided the score used for submission.

6.4 Kaggle Performance

- Final ensemble score : 0.88
- Rank: 125

7. Conclusion & Future Work

7.1 Key Learnings

- Noise injection from ESC-50, when done with fold discipline and curriculum, significantly improves robustness in noisy genre classification tasks.
- Pretrained audio Transformers fine-tuned with careful optimization (warmup, cosine schedule, layer-wise LR decay) provide a strong baseline even on relatively small datasets.
- Ensembling diverse architectures (Transformer, CNN hybrid, RNN) and multiple seeds is critical to push from high-0.90/0.95 into the >0.95 regime.

7.2 Challenges Faced

- Balancing noise strength: too much aggressive augmentation degraded convergence; curriculum and SNR scheduling were essential.
- Avoiding data leakage: ESC-50 fold discipline and deterministic validation noise were required to trust validation metrics.
- Overfitting: the dataset is relatively small; we had to aggressively use cross-validation, regularization, and augmentation to generalize well.

7.3 Future Work

- Explore larger or more specialized audio Transformers (e.g., AST variants or domain-specific models) with more advanced fine-tuning schemes.
- Investigate pseudo-labeling on unlabeled audio to further augment training data, with careful noise handling.
- Use more sophisticated ensembling (stacking, meta-learners) and per-class calibration.
- Incorporate multi-task learning with ESC-50 class prediction as an auxiliary task instead of only noise augmentation.
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8. References

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